Unit 4 · Outcome 1 · Climate Change

Responsible decision-making on climate change

KK13 VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 13 Responsible decision-making on climate change The whole chapter has been building to one hard question: now that we know what's coming, what should we actually do? This dot point is about the interconnections and tensions between the four forces that shape every real climate decision — stakeholder values, regulatory frameworks, scientific data and new technologies

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Responsible decision-making on climate change

UNIT 4 · OUTCOME 1 · KEY KNOWLEDGE 13 Responsible decision-making on climate change The whole chapter has been building to one hard question: now that we know what's coming, what should we actually do? This dot point is about the interconnections and tensions between the four forces that shape every real climate decision — stakeholder values, regulatory frameworks, scientific data and new technologies . Stakeholder values Regulatory frameworks Scientific data New technologies Throughout this chapter you've worked at the Cape Wirreanda Climate Observatory

01The four factors VCAA wants you to name

01 The four factors VCAA wants you to name Every climate decision is pulled on by four forces. The study design names them exactly — learn these four words, because a question can simply ask you to apply them: ① Stakeholder values What people care about The beliefs and priorities of everyone affected — homes, jobs, Country, nature, future generations. Maps onto the value systems from U3·O1 (anthropocentric → ecocentric). ② Regulatory frameworks The rules of the game Treaties, laws, targets and planning rules that permit, require or forbid options — from

Factor	At Pellan Bay it looks like…
Stakeholder values	Residents fear losing homes; farmers fear saltwater on paddocks; Traditional Owners hold cultural responsibility for sea-Country and the saltmarsh; BirdLife wants the shorebird habitat protected; young people ask about their coastline in 2070.
Regulatory frameworks	Paris Agreement & Ramsar Convention (international); Climate Change Act 2022 (Cth) net-zero-2050 & EPBC Act 1999 protecting the Ramsar marsh (federal); Climate Action Act 2017 (Vic) net-zero-2045, the Marine and Coastal Act 2018 and climate-aware planning scheme (state/local).
Scientific data	The observatory's 30-year tide record; IPCC AR6 sea-level projections with confidence ratings; CSIRO/BoM State of the Climate and the Victorian Climate Science Report 2024 ; council coastal-hazard inundation maps.
New technologies	A proposed Wirreanda offshore wind zone (aligned with Victoria's legislated 2 GW-by-2032 offshore-wind target), grid-scale batteries, nature-based "living shoreline" design, and saltmarsh blue-carbon restoration.

02Interconnections and tensions

02 Interconnections and tensions This is the heart of the dot point — and the single most common place students lose marks. The four factors don't sit in separate boxes. They pull on each other . The exam word "interconnections and tensions" is really two ideas: Interconnection Factors reinforce each other One factor strengthens, justifies or enables another. Example: scientific data on sea-level rise justifies the regulatory target, which in turn funds and mandates new offshore-wind technology. Tension Factors pull in opposite directions One factor bloc

03The Pellan Bay decision arena

✎ **P·E·L WRITING**

03 The Pellan Bay decision arena The Shire has three live proposals. Select each one to see how it scores on the four factors, what it trades off , and whether it tackles the cause (mitigation) or the consequence (adaptation) of climate change. There is no perfect option — that's the point. OPTION A Build a seawall A 2.3 km rock revetment and raised levee fronting the township. Holds the line, protects homes and the road now. ADAPTATION · hard protection OPTION B Managed retreat + living shoreline Voluntary buy-back of the most-exposed homes, relocate th

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

04Stakeholders and the values they hold

04 Stakeholders and the values they hold "Stakeholder values" is factor ①, and it connects straight back to the value systems from Unit 3 (anthropocentrism, biocentrism, ecocentrism, technocentrism). Different stakeholders weigh the same evidence completely differently — which is exactly why decisions create tension. Drag each Pellan Bay stakeholder onto the value lens that best captures their dominant priority. (On a phone, tap a stakeholder then tap a bin.) Anthropocentric / economic Value = use & benefit to people Ecocentric Whole ecosystem has intrin

05What turns a decision into a responsible one

05 What turns a decision into a responsible one VCAA uses the word "responsible" deliberately. A decision isn't responsible just because a choice was made. Use this checklist as the structure for any "evaluate / justify the decision" answer: Evidence-based Grounded in the best available scientific data , and honest about its confidence level and uncertainty — not cherry-picked. Precautionary Applies the precautionary principle : where there's a risk of serious or irreversible harm, lack of full certainty is not a reason to delay protective action. Equita

06Command-word decoder

◈ **COMMAND WORDS**

06 Command-word decoder This dot point is almost always tested with high-order command words. Read what each is really asking before you write — tap a word: Tap a command word Each one demands a different shape of answer. "Identify" wants a word; "evaluate" wants a weighed judgement with reasons. Matching your structure to the command word is worth easy marks.

07P·E·L writing trainer

✎ P·E·L WRITING

07 P·E·L writing trainer For the longer "analyse / evaluate" responses, build each paragraph as Point → Evidence → Link . Here's a worked P·E·L for a 4-mark "analyse the tension between two factors" question: POINT State the interaction directly Scientific data and stakeholder values are in tension over the Pellan Bay seawall. EVIDENCE Cite the specific factors The observatory's tide record and IPCC AR6 projections (high confidence) justify acting now on inundation, yet many residents place greater value on keeping their homes and the cost burden than on

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08Six exam traps

⚠ EXAM TRAPS

08 Six exam traps Trap 1 · Listing, not linking Naming the four factors separately answers a different (lower-mark) question. "Interconnections and tensions" demands you show how factors act on each other . Trap 2 · Mitigation ≠ adaptation The wind farm (mitigation) won't stop water already locked in; the seawall (adaptation) won't slow warming. Don't swap them. Trap 3 · Ignoring uncertainty "Evaluate" answers should reference confidence ratings and the precautionary principle, not treat projections as fixed fact. Trap 4 · Treating "stakeholders" as one

09Practice questions

★ PRACTICE & SELF-MARK

09 Practice questions Five VCAA-style questions · 20 marks total. Attempt each before revealing the responses. Mark yourself honestly — your score feeds the dock in the corner. my marks Reveal all responses

10One-screen cheat sheet

✦ RECAP / CHEAT SHEET

10 One-screen cheat sheet KK13 · Responsible decision-making — climate change The four factors ① Stakeholder values ② Regulatory frameworks ③ Scientific data ④ New technologies Interconnection Factors reinforce/enable each other (science justifies regulation; regulation funds technology). Tension Factors clash (Ramsar/EPBC rule blocks the seawall tech; science says act, residents' values resist). Mitigation Reduce the cause — cut emissions (offshore wind, decarbonisation). Adaptation Manage the consequence — seawall (hard) or managed retreat + living sho

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.